Obesity Risk Prediction: A Machine Learning Approach

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Executive Summary

Obesity is a major global health crisis, particularly affecting low-income populations in developing countries. Despite ongoing public health interventions, obesity rates continue to rise, driven by urbanization, dietary habits, and lifestyle changes. This study addresses the urgent need for an accurate and scalable obesity prediction model tailored to developing nations, where healthcare disparities further exacerbate the issue.

Our research aims to develop a machine learning-based obesity prediction model specifically designed for relatively low-income populations. Our study leverages a dataset containing 17 attributes and 2,111 records, categorizing individuals into seven distinct obesity levels. Exploratory Data Analysis (EDA) revealed that key predictors of obesity include weight, height, diet, family history, and physical activity levels. Feature engineering and data standardization were applied to optimize model performance.

A comparative analysis of machine learning models demonstrated that Multi-Layer Perceptron (MLP) achieved the highest accuracy (97.64%), outperforming other approaches. Support Vector Machines (SVM) and Gradient Boosting also exhibited strong performance, with accuracy scores of 96.69% and 96.22%, respectively. While Deep Neural Networks (DNN) have the potential to excel in large-scale datasets, they underperformed in this study due to data constraints and potential overfitting.

Despite achieving high accuracy, the study identified several limitations. The dataset primarily focuses on dietary and physical activity variables while lacking socioeconomic and genetic factors that significantly influence obesity. Additionally, model generalizability remains a concern, as obesity trends vary by region, age, and economic status. To improve

predictive power, future research should integrate broader datasets, including behavioral and socioeconomic variables, and leverage advanced model optimization techniques such as Bayesian tuning and ensemble learning.

From a policy perspective, data-driven interventions are critical to combating obesity in developing nations. Machine learning can enhance early detection and enable personalized health recommendations. However, technology alone cannot resolve the crisis. A holistic approach combining AI-powered health monitoring, improved public health policies, and behavior-focused interventions is essential to mitigating obesity rates. Governments should strengthen regulatory measures such as sugar taxes, food labeling reforms, and community-based exercise programs to foster healthier lifestyles.

1.0 Problem Statement

Obesity is a growing global health crisis, with increasing prevalence in low- and middle-income countries due to economic inequality, lifestyle changes, and lack of effective interventions. Traditional statistical methods have been used to estimate obesity risk, but they often fail to capture high-dimensional relationships among socioeconomic, behavioral, and genetic factors. Recent advances in machine learning (ML) offer promising solutions, demonstrating higher predictive accuracy. However, existing ML-based models often rely on datasets from developed nations, making them less generalizable to diverse populations. Additionally, there is a gap between ML-driven obesity predictions and real-world public health interventions, limiting their impact on policy-making. Addressing these challenges requires an integrated approach that improves prediction accuracy, interpretability, and applicability in policy-driven healthcare strategies.

2.0 Research objectives

This research aims to improve obesity prediction models by leveraging machine learning techniques to enhance classification accuracy, model interpretability, and real-world applicability. Specifically, the study seeks to: Develop a machine learning-based obesity prediction model tailored for developing countries, using datasets from Latin America to capture region-specific obesity trends and risk factors; integrate obesity prediction with public health interventions, designing a framework that aligns ML-driven risk assessments with policy recommendations and targeted intervention strategies; evaluate the interpretability and real-world applicability of different machine learning models, balancing predictive performance with transparency and ease of implementation; Assess the feasibility of deploying ML-driven obesity prediction tools in real-world healthcare settings, evaluating factors such as data availability and computational efficiency.

3.0 Literature Review

Obesity is a growing public health crisis, particularly in developing countries, where rising economic inequality exacerbates the issue. Defined as a body mass index (BMI) of 30 kg/m^2 or higher, obesity is associated with a higher risk of chronic diseases, including type 2 diabetes, cardiovascular disease, and metabolic disorders (Ferreira et al. 2024). Although machine learning (ML) techniques have been increasingly applied to predict obesity levels, existing models often rely on data from developed nations, limiting their effectiveness in low-income regions (Lucena et al. 2024). Furthermore, the integration of ML-based predictions with public health interventions remains insufficient, hindering the translation of computational insights into actionable policies (Ferreira et al. 2024). Addressing these challenges is critical for improving obesity prevention and management strategies, particularly in resource-constrained settings.

A study develops an obesity classification dataset using data from Mexico, Peru, and Colombia, incorporating 17 demographic, physical, and behavioral attributes (Mendoza Palechor and de la Hoz Manotas 2019). The dataset is designed to support machine learning applications, addressing the issue of class imbalance using Synthetic Minority Over-Sampling Technique (SMOTE). While this dataset provides a strong foundation for obesity prediction, it primarily focuses on dietary and physical activity variables, overlooking socioeconomic and genetic influences, which play a key role in obesity risk.

Another study applies the Box–Cox Power Exponential (BCPE) model to national health survey data from Mexico, Colombia, and Peru, highlighting the limitations of traditional BMI prediction models in capturing extreme values (Lucena et al. 2024). By adjusting for the skewed nature of BMI distributions, this model provides better estimates of obesity prevalence. However, while the BCPE model offers statistical advantages, it does not incorporate high-dimensional behavioral and socioeconomic factors, which machine learning techniques can better analyze.

A demographic study finds significant differences in obesity trends based on age and gender, using health survey data from Mexico, Colombia, and Peru (Yamada et al. 2019). The research reveals that BMI distributions among women are more right-skewed, indicating higher obesity prevalence compared to men. Additionally, obesity rates among men aged 35–39 in Peru increased by 21 percentage points over five years, suggesting that demographic-specific prevention strategies are necessary. However, most ML-based obesity prediction models do not incorporate gender- and age-specific risk factors, limiting their effectiveness in designing targeted interventions.

Beyond individual behaviors, environmental and socioeconomic determinants significantly impact obesity prevalence. A study emphasizes that rapid urbanization, economic inequality, and industrialized food production have increased the consumption of ultra-processed foods and sugary beverages, contributing to higher obesity rates in Latin America (Ferreira et al. 2024). The research further highlights how limited healthcare access and weak regulatory policies disproportionately affect low-income populations, making policy-driven interventions essential for effective obesity management.

Recent studies evaluate the performance of various machine learning algorithms in obesity prediction, comparing Random Forest, Support Vector Machines (SVM), Gradient Boosting, and Neural Networks (Ponte de Lucena et al. 2024). Findings indicate that Gradient Boosting and Neural Networks outperform traditional models, achieving higher accuracy in predicting obesity risk. However, results suggest that models trained solely on dietary or physical activity data perform less effectively than models integrating both nutritional and behavioral risk factors, reinforcing the need for multifactorial prediction approaches.

Despite advancements in ML-based obesity classification, imbalanced class distributions remain a key challenge. A study introduces SMOTE as a solution to improve classification performance for underrepresented obesity categories, showing that oversampling significantly enhances model recall (Chawla et al. 2002). However, researchers caution that SMOTE alone does not fully resolve class imbalance, as it may introduce class overlap near decision boundaries, reducing classification precision. Future research should explore cost-sensitive learning and generative adversarial networks (GANs) to improve obesity classification.

Another concern in ML-driven obesity research is model transparency and interpretability. A study compares different ML-based obesity classification models, analyzing the trade-off between accuracy and explainability (DeGregory et al. 2018). The research suggests that while Deep Neural Networks (DNNs) provide high accuracy, simpler models such as Decision Trees and Random Forests offer greater transparency, making them more suitable for real-world public health applications.

Machine learning has also been applied to predict long-term obesity-related health complications. A study develops predictive models for obesity-related diseases, using electronic health records from over 400,000 patients (Turchin et al. 2024). By applying Lasso-Cox regression and Random Survival Forests (RSF), researchers accurately predict nine long-term health risks associated with obesity, demonstrating that ML can enhance early detection and clinical intervention strategies.

The economic impact of obesity has been extensively studied. A cost-effectiveness analysis evaluates the financial burden of obesity prevention, showing that preventative strategies lower short-term healthcare costs by reducing the incidence of chronic diseases such as type 2 diabetes and cardiovascular conditions (Rappange et al. 2009). However, the research highlights a policy dilemma, as individuals who avoid obesity tend to live longer, leading to increased long-term healthcare costs. This paradox underscores the need for balancing cost-effectiveness with sustainable healthcare planning.

A systematic literature review synthesizes machine learning applications in obesity prediction, analyzing over 90 studies (Safaei et al. 2021). The findings conclude that ML-driven early detection significantly improves obesity prevention efforts, particularly when integrated with electronic health records (EHRs) and wearable technology. However, data privacy concerns, computational constraints, and model biases remain key challenges preventing large-scale implementation.

The literature review highlights the growing role of machine learning in obesity research while addressing ongoing challenges in data inclusivity, model transparency, and policy integration. Despite high classification accuracy, many ML models depend on controlled datasets, restricting their applicability in resource-limited healthcare settings. Although progress has been made in explainable AI and cost-effective obesity prevention, ML-driven insights remain underutilized in public health decision-making. Future studies should focus on developing ML-based obesity prediction frameworks that align with public health policies, ensuring AI-driven findings effectively support prevention and intervention efforts.

4.0 Exploratory Data Analysis

4.1 Data Overview

The dataset comprises numerical and categorical features that influence obesity levels, with "Obesity" serving as the target variable. The numerical features include Age, Height, Weight, and various lifestyle factors, such as physical activity (FAF) and water consumption (CH2O). The categorical features encompass attributes like family history of obesity, dietary habits, and transportation modes.

4.2 Data Visualization and Distribution Analysis

4.2.1 Numerical Feature Distributions

Histograms were plotted for all numerical variables to assess their distributions. Key observations include that most individuals fall within the 18–30 age range, indicating a young population distribution. Height follows a near-normal distribution but exhibits slight skewness towards shorter individuals. Weight displays considerable variation across obesity classifications. Several dietary and lifestyle-related numerical variables, such as FCVC, NCP, CH2O, FAF, and TUE, show distinct peaks, suggesting that they behave more like categorical variables despite being numerical in nature.

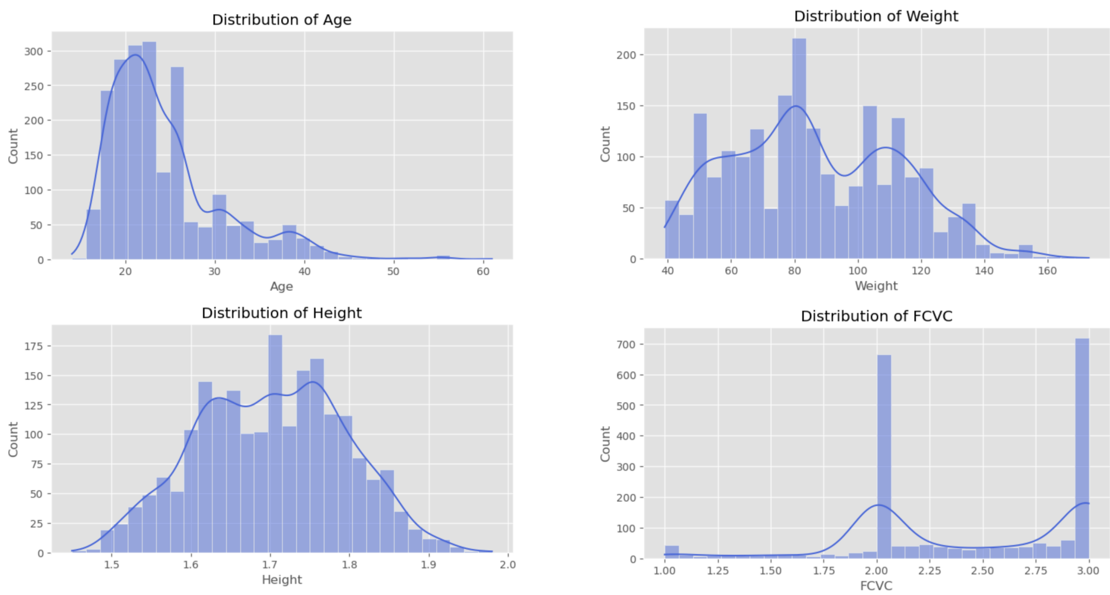

Figure 1: Distribution of Numerical Variables

4.2.2 Categorical Feature Distributions

Count plots were generated for categorical features, revealing various trends. A significant proportion of obese individuals have a family history of obesity, highlighting a genetic predisposition. Many respondents frequently consume high-caloric foods and snacks between meals, reinforcing the role of diet in obesity. Additionally, individuals with higher obesity levels tend to consume alcohol more frequently. The mode of transportation also plays a role, with most individuals using public or private transport rather than engaging in active commuting.

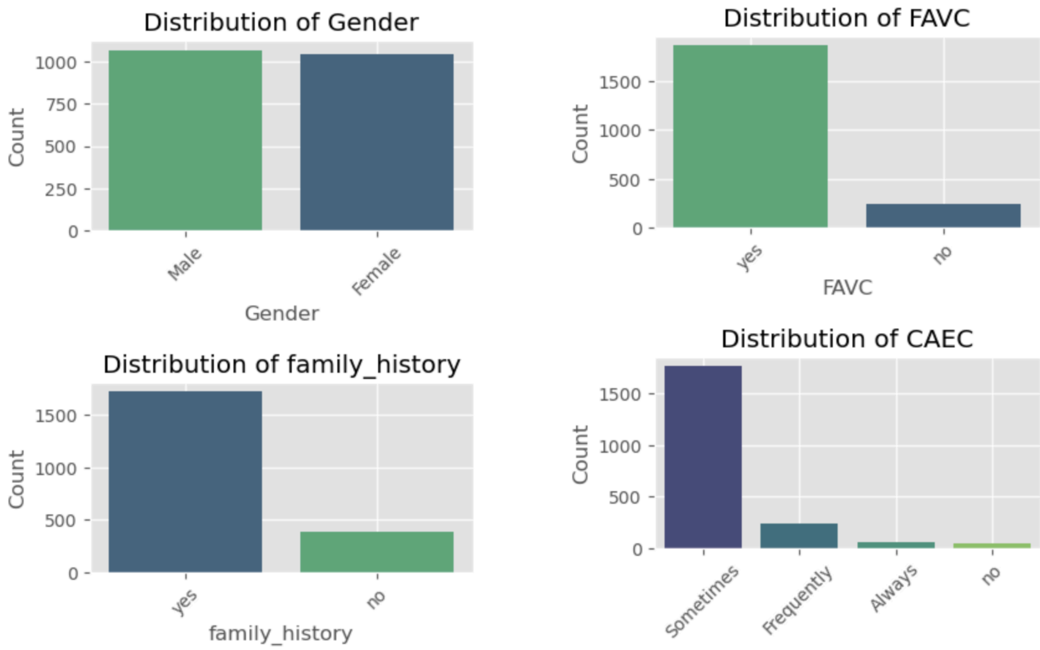

Figure 2: Distribution of Categorical Variables

4.3 Feature Interactions and Relationships

4.3.1 Boxplots: Obesity vs. Numerical Features

Boxplots illustrated the relationship between obesity levels and key numerical variables. Younger individuals tend to exhibit higher obesity levels, while height does not display a clear pattern, although lower height appears more common in higher obesity categories. Weight increases progressively with higher obesity classifications, confirming its expected correlation with obesity.

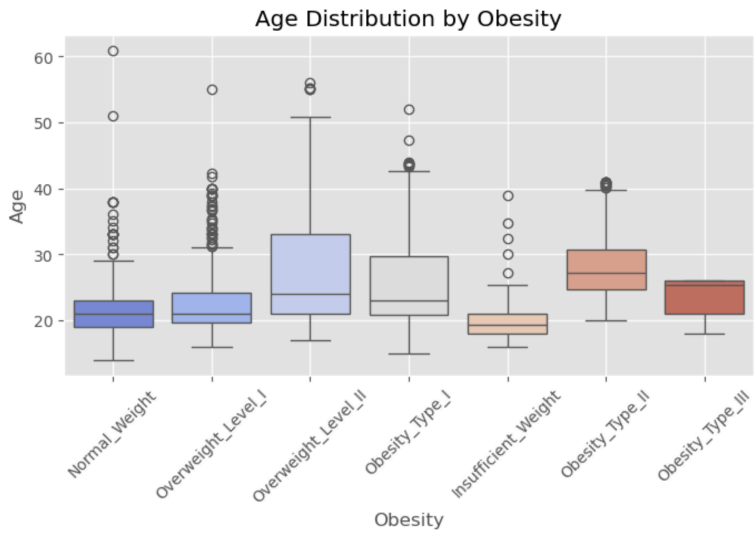

Figure 3:Age Distribution by Obesity

4.3.2 Correlation Matrix

A heatmap of feature correlations highlighted significant relationships:

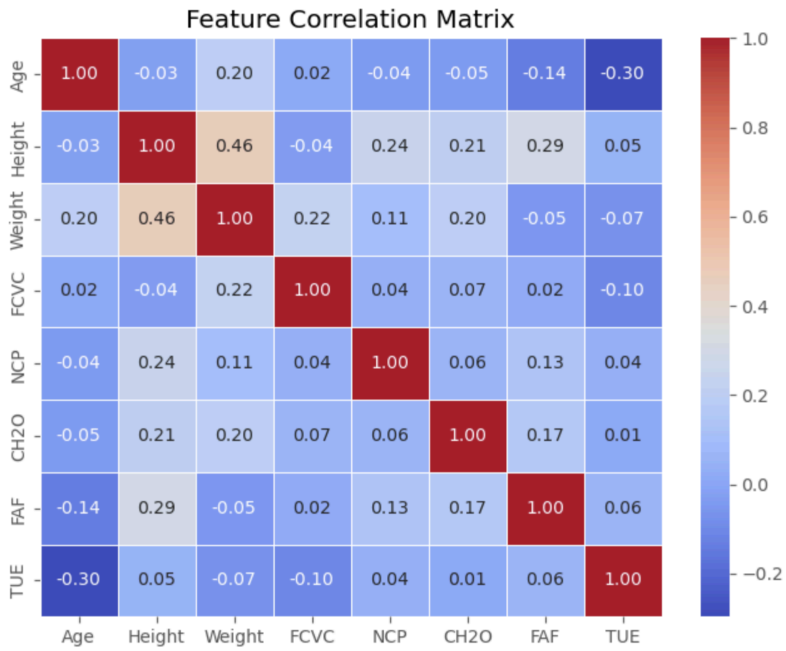

Figure 4: Feature Correlation Matrix

Through the correlation matrix, we observed that height and weight are positively correlated; however, weight alone is not the sole determinant of obesity. Physical activity (FAF) is negatively correlated with obesity, reinforcing the connection between a sedentary lifestyle and increased obesity risk. Water consumption (CH2O) shows a positive correlation with normal-weight individuals, suggesting a potential beneficial role of hydration in weight management. Technology usage (TUE) has a weak positive correlation with obesity, indicating a possible association between screen time and obesity levels.

4.3.3 Pairplot Analysis

A pairplot was utilized to explore multi-variable relationships and assess feature separability in the dataset. Among all variables, weight emerged as the strongest predictor of obesity levels. Additionally, individuals who maintain healthier habits, such as frequent vegetable consumption and adequate water intake, generally exhibit lower obesity levels. In contrast, prolonged screen time and insufficient physical activity are associated with an increased risk of obesity. These findings highlight the significant impact of lifestyle choices on obesity and underscore the importance of promoting healthy habits to mitigate obesity risks.

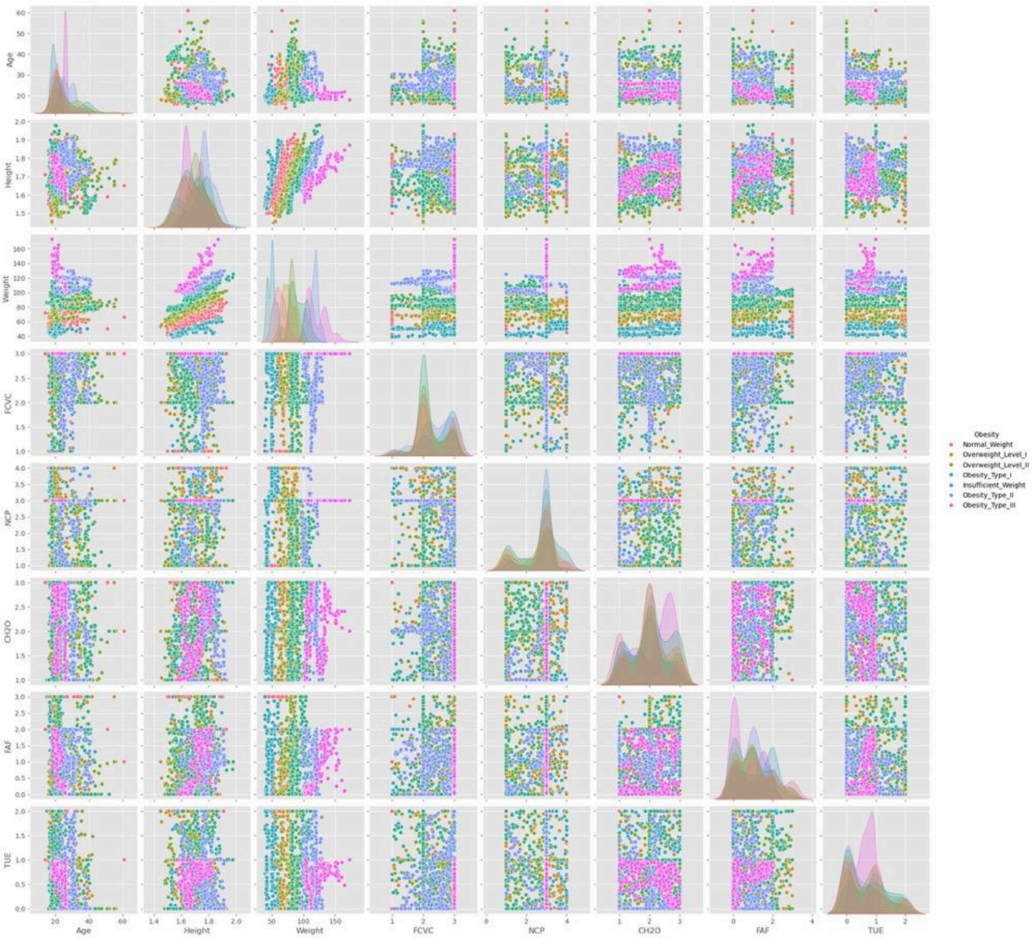

Figure 5: Pairplot

4.4 Key Insights and Takeaways

Diet and physical activity emerge as primary drivers of obesity, with frequent consumption of high-caloric foods (FAVC) and low physical activity (FAF) being the most influential factors. Most individuals do not actively monitor their calorie intake (SCC), suggesting potential intervention points for obesity prevention. Sedentary behavior (TUE) and genetic predisposition (family_history) contribute to obesity but with varying degrees of influence. Additionally, class imbalance in obesity categories may necessitate resampling techniques for model training to ensure robust predictions.

5.0 Data Preparation/Feature Engineering

5.1 Handling Missing Values

In order to deal with missing data, we used the SimpleImputer class from scikit-learn with the strategy parameter set to 'median'. The reason for using this method is because it replaces missing numerical values with the median of the respective feature, something that's better than using the mean because the median is less affected by extreme value instances (outliers) distorting the distribution. We filled in missing values so that we don't lose any information as the records were incomplete and all variables were made ready for further analysis.

5.2 Encoding Categorical Variables

Since the above, the machine learning algorithms need numerical inputs so we encoded all the categorical variables using LabelEncoder. For a feature like gender, each unique category in the feature (Male, Female) is converted to a unique integer value (Male = 0, Female = 1). The target variable Obesity was also encoded using this encoding and then converted into a numeric format. In this step, the data required to be compatible with the machine learning algorithms as well as the categorical relationships were intact. We systematically encoded these variables, so that the algorithms could work with the data in an effective way and without having the algorithms to interpret differently and to introduce inconsistencies.

5.3 Create New Feature

To extract important predictors of obesity in an interpretable and more meaningful fashion, Obesity Risk was engineered as these features used various combinations of the key predictors. We integrated these derived variables into the dataset and thereby improved the overall quality of the dataset, and fed into richer, relevant information about which the model could make accurate predictions.

$$\text{Obesity\_Risk} = \text{family\_history} \times \text{FAVC}$$

where:

- family_history indicates whether obesity is prevalent in the individual's family (binary: 0 or 1).
- FAVC represents whether the individual frequently consumes high-caloric foods (binary: 0 or 1).

5.4 Scaling Numerical Features

Features were then standardized to numerical scale in order for all of the predictors to be operating on similar scales. In order to transform the numerical variable to have a mean of 0 and a standard deviation of 1, we used the StandardScaler from scikit-learn. This standardization process decreases the impact of magnitude difference among variables. It also prevents features with large values from dominating the learning process in machine learning models like logistic regression and support vector machines which are sensitive to scale.

For those models that rely on some sort of distance metric between data points, like KNN or clustering techniques, this was a very critical step.

5.5 Feature Selection

To enhance model performance and reduce dimensionality, we applied a systematic feature selection process using three different methods: Recursive Feature Elimination (RFE), SelectKBest (ANOVA F-test), and LassoCV (L1 regularization). These methods were employed to identify the most relevant features for predicting the target variable.

Recursive Feature Elimination (RFE): We utilized logistic regression as the base estimator to iteratively remove the least important features, ultimately selecting the top 10 most informative ones. This method ensures that redundant or less relevant features are excluded based on their contribution to model performance.

SelectKBest (ANOVA F-test): A univariate feature selection approach was applied, where each feature was evaluated based on its statistical significance in relation to the target variable using the ANOVA F-test. The top 10 features with the highest F-scores were retained.

LassoCV (L1 Regularization): We employed Lasso regression with five-fold cross-validation to determine the optimal regularization parameter. Features with nonzero coefficients were selected, as Lasso inherently shrinks the coefficients of less important features to zero, thus performing feature selection automatically.

To determine the final set of features, we took the intersection of the selected features from all three methods to ensure robustness. If fewer than 10 features were retained, we incorporated additional features from the union of the three methods to maintain a meaningful feature set. This hybrid selection approach balances statistical significance, predictive power, and model interpretability, ensuring the most influential features are used in the final model.

5.6 Train-Test Split

For the reliable evaluation of the model we divided the dataset into training and testing sets with the 80:20 ratio. The split was done using the scikit-learn train_test_split function. This process stratified the target variable Obesity, to assure equal class distribution across these two subsets of the overall data. For imbalanced datasets, stratification plays an essential role, to help prevent skewed performance metrics and help guarantee that the model learns representative patterns for all classes. The machine learning model was trained by using a training set and final performance evaluation was done on a testing set that minimized the risk of overfitting.

5.7 Summary

A clean, informative and ready to go modeling dataset was prepared through our approach towards the data preparation and feature engineering. We enhanced the dataset by imputing missing values, encoding categorical data and scaling numerical features making the dataset more compatible with the machine learning algorithms. Explanatory variables created in the process are mediated through producing Obesity Risk for medical relevance, and interpretable variables, while Feature Selection guarantees removal of predictors not having an impact on our model. Last, we ensured a fair and unbiased evaluation of the model by a stratified train-test split, a ground for robust prediction.

6.0 Methodology & Tools

In this project, we employed a comprehensive Machine Learning pipeline to predict obesity levels from various demographic and behavioral features. Initially, we performed thorough exploratory data analysis utilizing Python libraries such as pandas, numpy, matplotlib, seaborn, and plotly to understand distributions, detect outliers, and visualize correlations among variables clearly.

For data preprocessing, we leveraged SimpleImputer for handling missing values and applied LabelEncoder and StandardScaler to encode categorical variables and standardize numerical features, respectively. Feature selection was executed through robust approaches, including Recursive Feature Elimination, SelectKBest, and regularization methods such as

Lasso regression, ensuring that we retained only significant predictors for the modeling phase.

Each model was implemented using Python’s scikit-learn library, while hyperparameter optimization was systematically performed using GridSearchCV to ensure optimal predictive performance. To evaluate model robustness and avoid overfitting, we applied K-fold cross-validation during the training process.The modeling was executed in a local development environment powered by Anaconda, running on a standard CPU without GPU acceleration.

7.0 Model Selection, Training, and Evaluation

7.1 Model 1: Random Forest Model

In the first modeling approach, a Random Forest Classifier was implemented. This model was selected because of its capability to handle complex, nonlinear relationships, manage high-dimensional data effectively, and provide robust predictions even in the presence of noise or correlated predictors. Additionally, Random Forests have built-in feature importance measures, making it easier to understand which variables most significantly influence obesity levels.

Initially, a baseline model was developed without tuning, resulting in an accuracy of approximately 93.14%.To ensure model stability and reliability, 5-fold cross-validation was conducted, achieving an average accuracy of around 93.90%, indicating the model’s generalization capability beyond the initial test set.

Recognizing potential room for improvement, hyperparameter tuning was performed using a systematic GridSearchCV approach. Multiple hyperparameters were tested systematically, including the number of trees (n_estimators), maximum depth of trees (max_depth), minimum samples required per leaf node (min_samples_leaf), minimum samples required for splits (min_samples_split), and the usage of bootstrapping. A thorough grid search combined with 5-fold cross-validation was performed to determine the optimal combination. The best parameters identified included 50 trees without bootstrapping, a maximum depth of 20, and carefully adjusted minimum sample requirements, collectively enhancing the prediction accuracy and overall robustness of the model.

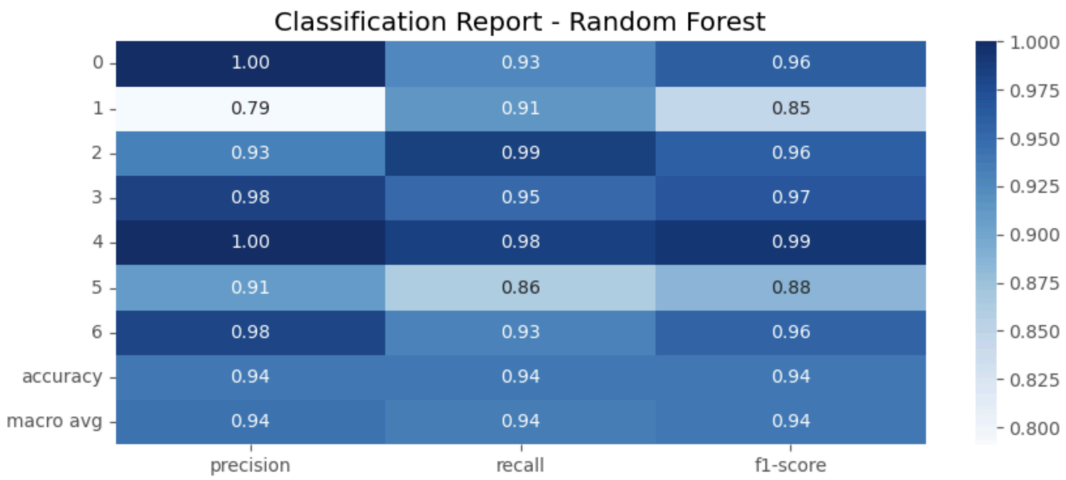

Figure 6: Classification Report of Model 1

The classification report shows an overall accuracy of approximately 94%, demonstrating the model's strong capability in predicting obesity levels. The model performs particularly well in terms of precision and recall for most classes, although Class 5 exhibited slightly lower recall (86%), suggesting the model has some difficulty identifying instances of this category correctly.

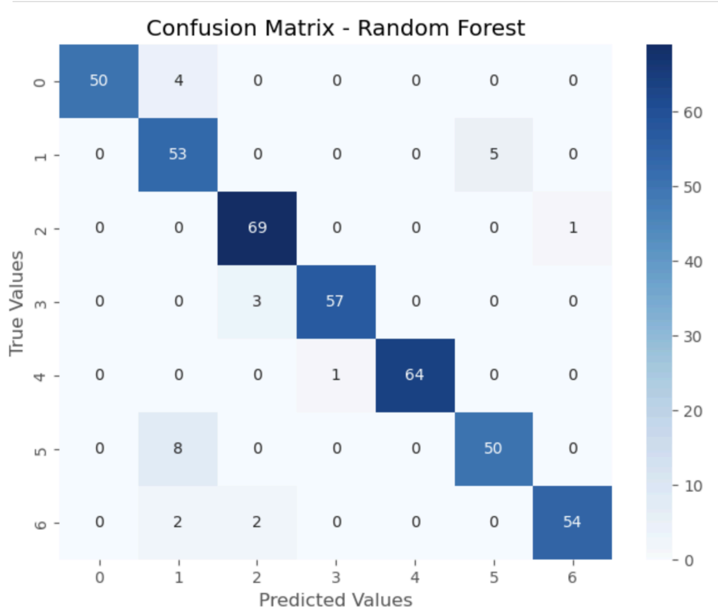

Figure 7: Confusion Matrix of Model 1

The confusion matrix indicates that most classes were predicted accurately, with only minor misclassifications observed, primarily between closely related obesity categories. This result underscores the model's ability to discriminate effectively between most classes, though it highlights specific categories where misclassification occurs more frequently, suggesting areas for further improvement.

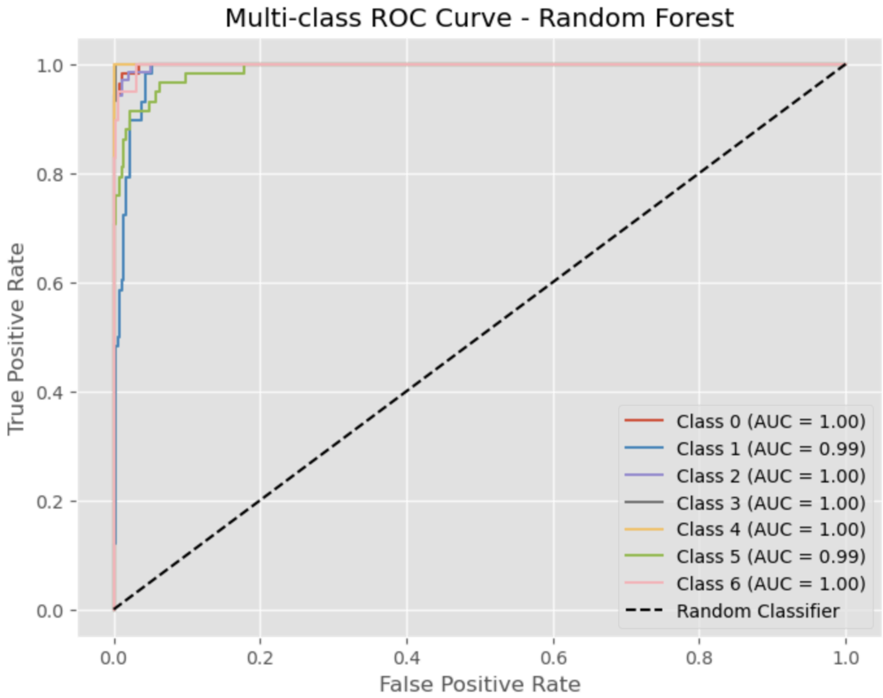

Figure 8: ROC Curve of Model 1

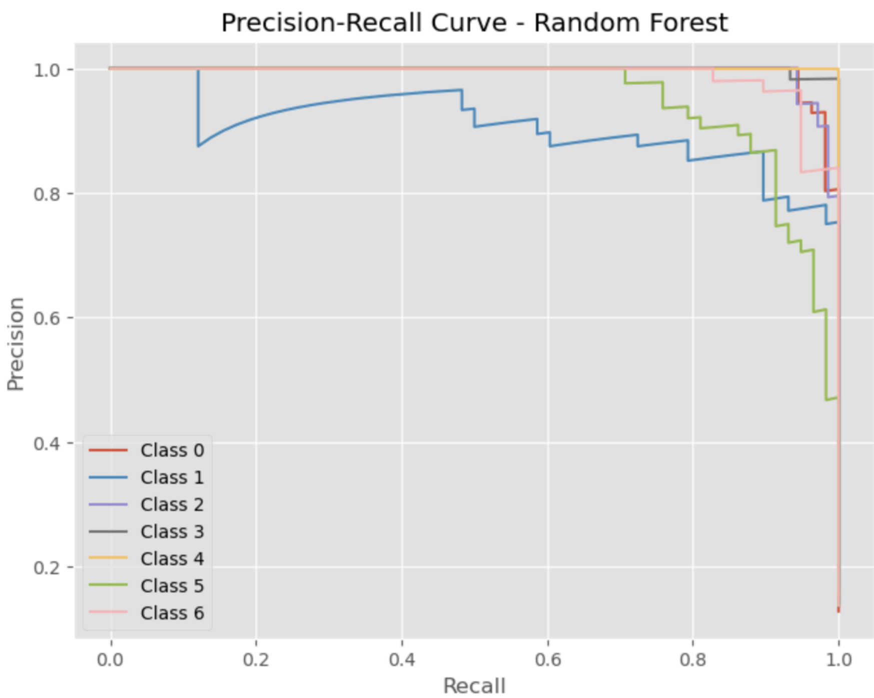

Figure 9: Precision-Recall Curve of Model 1

Moreover, the precision-recall curves and multi-class ROC curves provided additional validation of the classifier's performance. The ROC curves demonstrate near-perfect classification (AUC close to 1.00) for several classes, further confirming the robustness of the predictive model. Nevertheless, some classes exhibited lower precision or recall (e.g., Class 5), suggesting that while the model performs well generally, additional targeted tuning might be beneficial to enhance the prediction quality of specific classes.

The selected Random Forest model represents a good compromise between complexity and interpretability, efficiently predicting obesity classification from demographic and lifestyle data. Future improvements could focus on addressing class-specific prediction challenges, perhaps by adjusting sampling techniques or exploring additional modeling strategies.

7.2 Model 2: Gradient Boosting Model

The second model employed was a Gradient Boosting Classifier, selected primarily due to its strong predictive power and ability to effectively capture complex, nonlinear relationships and interactions among explanatory variables. Gradient Boosting algorithms excel in classification tasks by iteratively combining multiple weak models into a strong predictive framework, which is especially beneficial for challenging classification problems.

Initially, a baseline model with default hyperparameters was built, which achieved an accuracy of approximately 95.74%, serving as a strong initial benchmark. 5-fold cross-validation was conducted, achieving an average accuracy of around 95.26%, indicating the model's generalization capability beyond the initial test set.

To enhance predictive performance, hyperparameter tuning was conducted systematically using GridSearchCV in conjunction with 5-fold cross-validation. The

GridSearchCV method explored multiple critical hyperparameters, including the number of boosting stages (n_estimators), the maximum depth of each decision tree (max_depth), the minimum number of samples required to split internal nodes (min_samples_split), and the minimum samples needed for leaf nodes (min_samples_leaf).The best-performing model was identified with the following optimal parameters: a learning rate of 0.1, maximum tree depth of 5, 100 boosting stages, minimum samples per leaf set to 2, and minimum samples per split of 5.This optimized Gradient Boosting model notably enhanced prediction accuracy to 96.22%, demonstrating clear improvements over the baseline model and confirming the effectiveness of meticulous hyperparameter optimization.

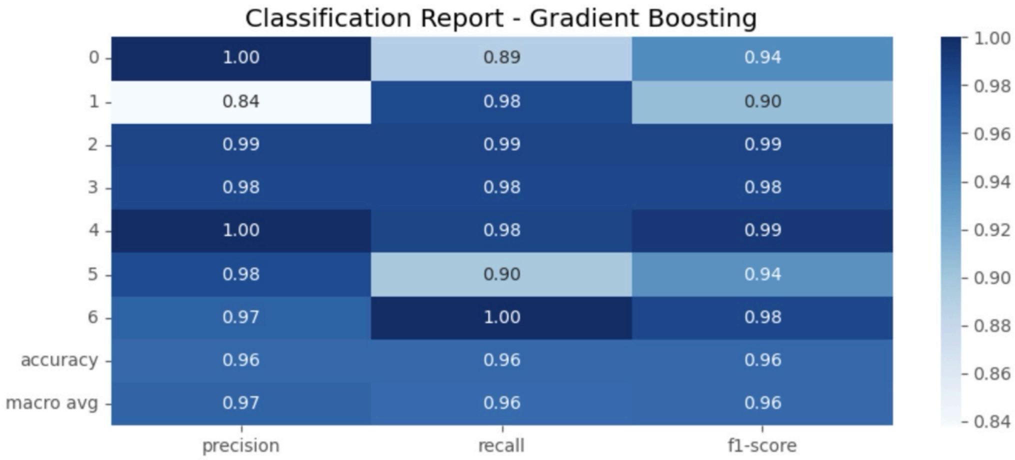

Figure 10: Classification Report of Model 2

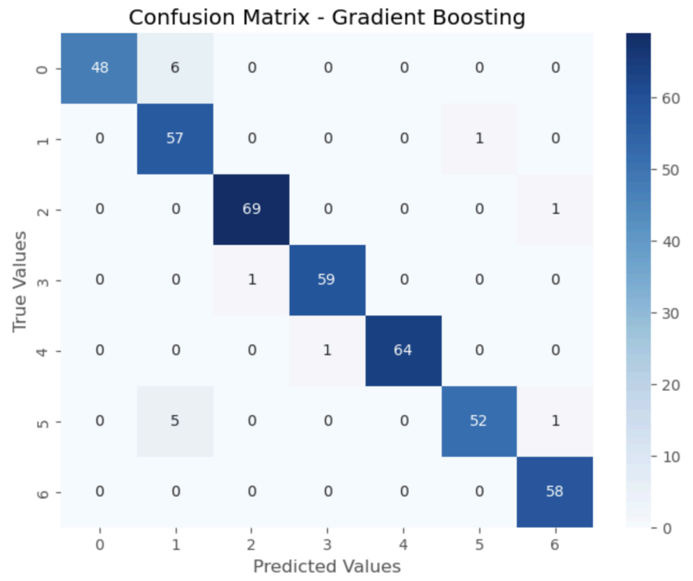

Figure 11: Confusion Matrix of Model 2

The confusion matrix clearly demonstrated high classification accuracy across all obesity classes, with only minor misclassifications. The precision, recall, and F1-score metrics from the classification report further confirmed high model reliability, consistently surpassing 95% accuracy across most classes.

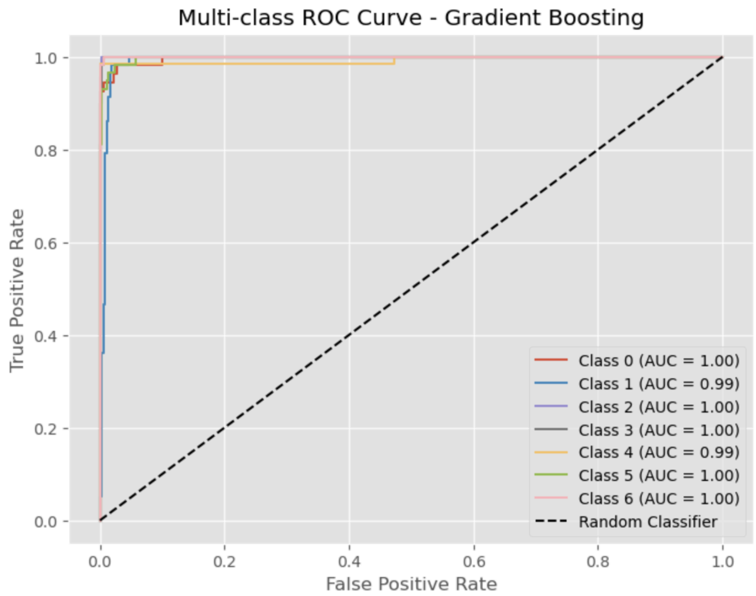

Figure 12: ROC Curve of Model 2

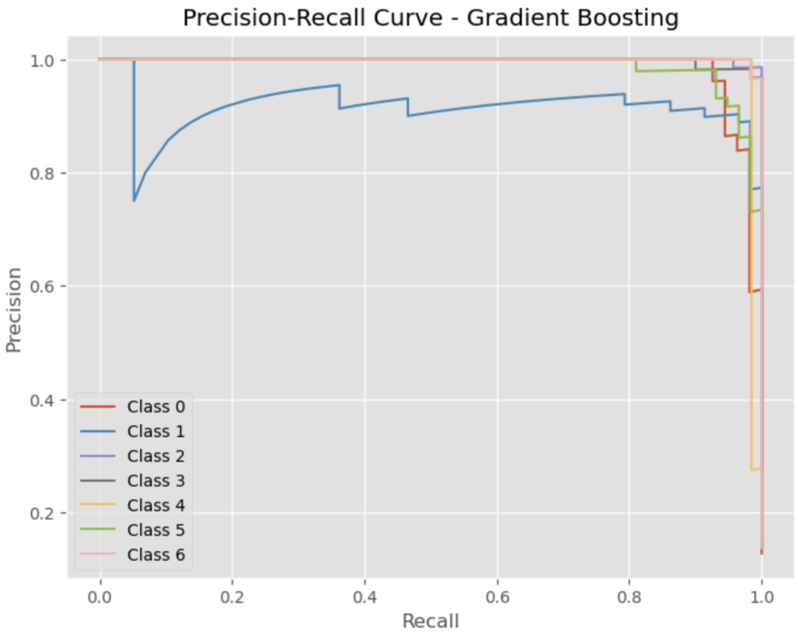

Figure 13: Precision-Recall Curve of Model 2

Additionally, multi-class ROC curves highlighted strong discriminatory capability, with areas under the curve close to or equal to 1.00 for almost all classes, signifying excellent overall predictive power. The Precision-Recall Curve provided valuable insights into precision-recall trade-offs, revealing robust class-specific performance with minor challenges in recall for certain classes. Specifically, minor weaknesses were observed in classes 1 and 5, suggesting areas that could benefit from targeted adjustments or additional feature engineering.

The decision to select the Gradient Boosting Classifier was justified by its proven predictive strength and its suitability for capturing complex, nonlinear data relationships. The rigorous hyperparameter tuning significantly enhanced the model's accuracy from baseline, indicating the value and necessity of detailed parameter exploration.

7.3 Model 3: Support Vector Machine Model

Model 3 utilized a Support Vector Machine (SVM), a robust classification algorithm particularly effective with datasets where clear margin separation between classes is desired. SVM was selected due to its ability to handle complex, high-dimensional feature spaces and its effectiveness in achieving strong generalization performance.

Initially, a baseline SVM model was constructed using the default parameters. The baseline model achieved an accuracy of approximately 92.67% on the test set, demonstrating already promising results before optimization. 5-fold cross-validation was conducted, achieving an average accuracy of around 91%.

To further enhance the model's predictive capability, a comprehensive hyperparameter tuning procedure was performed using GridSearchCV with 5-fold cross-validation. This exhaustive grid search explored various combinations of hyperparameters, specifically testing the penalty parameter C values, kernel types (linear, rbf, and polynomial degrees), and the kernel coefficient gamma. After evaluating multiple configurations, the optimal hyperparameters identified were a linear kernel, C=10, and gamma='scale', highlighting that a linear decision boundary provided the best trade-off between model complexity and predictive accuracy. Besides, the optimized SVM significantly improved performance, achieving an accuracy of 96.69%, surpassing the baseline by about 4%.

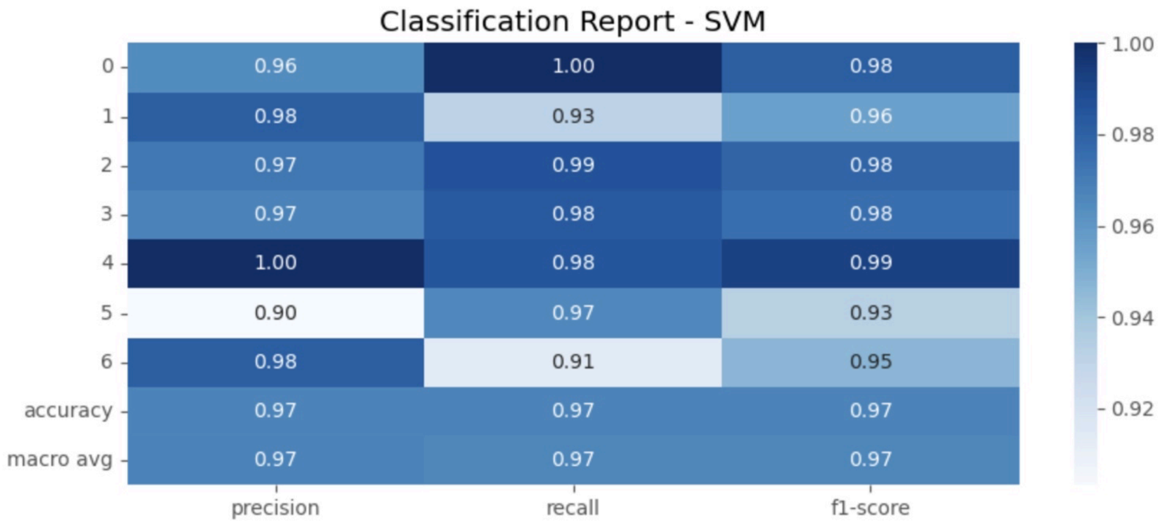

Figure 14: Classification Report of Model 3

Comprehensive evaluation through the provided classification report indicates that the optimized SVM achieved excellent precision, recall, and F1-scores across all obesity classes. Particularly notable were classes 0, 2, 3, and 4, which consistently scored high (around or above 0.97) across precision, recall, and F1-score metrics. However, class 1 and class 6 showed slightly lower recall values (0.93 and 0.91, respectively), suggesting some misclassification occurrences that deserve further examination. These lower recall scores imply that the model occasionally struggles to detect certain instances correctly in these categories.

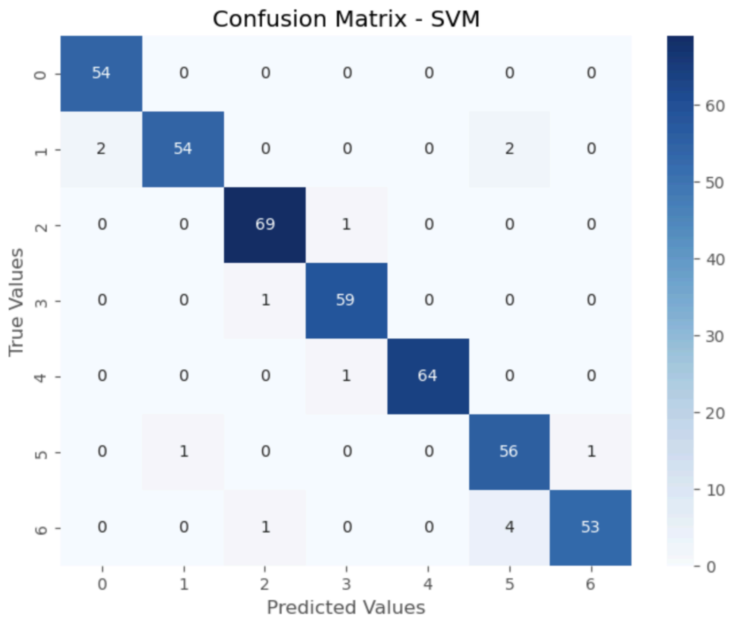

Figure 15: Confusion Matrix of Model 3

The confusion matrix demonstrated high accuracy and minimal misclassification across classes, although minor classification errors occurred, primarily within classes 5, indicating the need for further refinement. Precision and recall scores presented in the classification report were consistently high across all classes, though certain classes, notably class 5, exhibited slightly lower recall, highlighting targeted areas for further improvement.

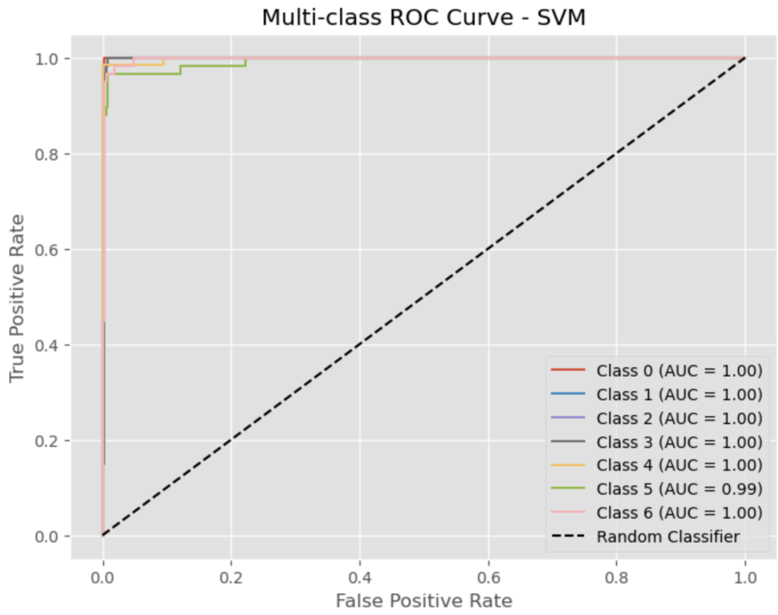

Figure 16: ROC Curve of Model 3

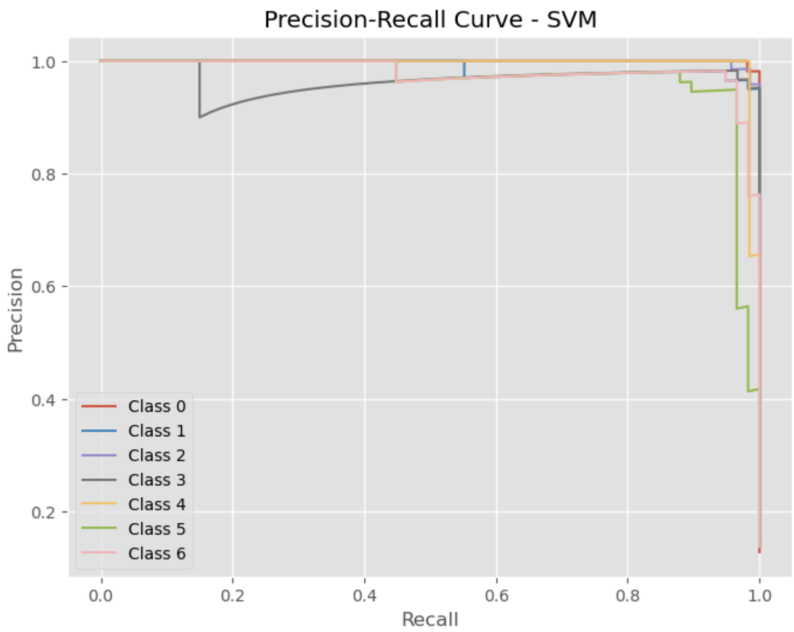

Figure 17: Precision-Recall Curve of Model 3

Moreover, the multi-class ROC curve showcased excellent overall discriminative power across classes, with the AUC approaching 1.00 for most classes. The Precision-Recall curve complemented this, providing insights into precision-recall trade-offs, further reinforcing the robustness of model performance across different classes.

Overall, the evaluation metrics, visualizations, and systematic hyperparameter tuning provided clear justification for selecting SVM as an effective predictive model, offering robust classification performance and reliability for predicting obesity levels.

7.4 Model 4: Neural Network Model

Model 4 implemented a feedforward Neural Network. This model was selected for its ability to capture complex, non-linear relationships in the data, offering flexibility beyond traditional linear models. Neural networks are particularly effective in multi-class classification problems where feature interactions are intricate and cannot be easily modeled by simpler algorithms.

A baseline neural network was initially constructed with default parameters, including a maximum of 300 iterations and a fixed random seed for reproducibility. This baseline model achieved an accuracy of 95.98% on the test set, providing a solid starting point for further refinement.

To improve performance, hyperparameter tuning was conducted through GridSearchCV with 5-fold cross-validation. The tuning process evaluated 108 combinations of hyperparameters, including the number of hidden layers, activation functions (relu, logistic), solvers (adam, lbfgs), regularization strength (alpha), and learning rate strategies. The optimal configuration consisted of two hidden layers with 100 neurons each, logistic activation, the Adam solver, an alpha value of 0.001, and a constant learning rate.

After tuning, the optimized neural network achieved a test accuracy of 97.64%, reflecting a notable improvement over the baseline and confirming the effectiveness of detailed hyperparameter optimization. This structured approach ensured the neural network was well-calibrated to the dataset's complexity, achieving both high accuracy and generalization.

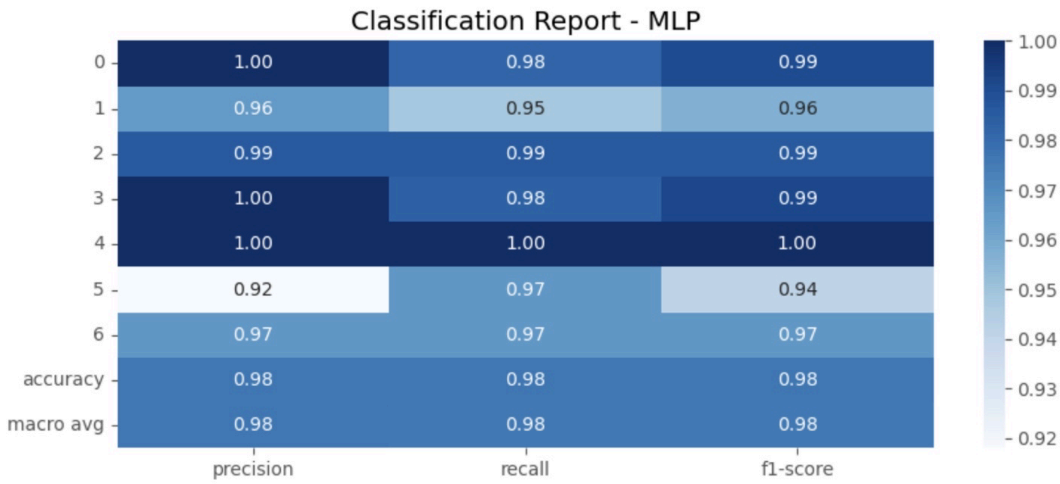

Figure 18: Classification of Model 4

The classification report reveals consistently high levels of precision, recall, and F1-scores across all obesity classes. Specifically, classes 0, 2, 3, and 4 achieved near-perfect or perfect scores, while classes 1, 5, and 6 maintained strong but slightly lower metrics, particularly in recall and precision for class 5. The overall accuracy reached 98%, and the macro-averaged F1-score was also 0.98, indicating balanced model performance across both frequent and less frequent classes.

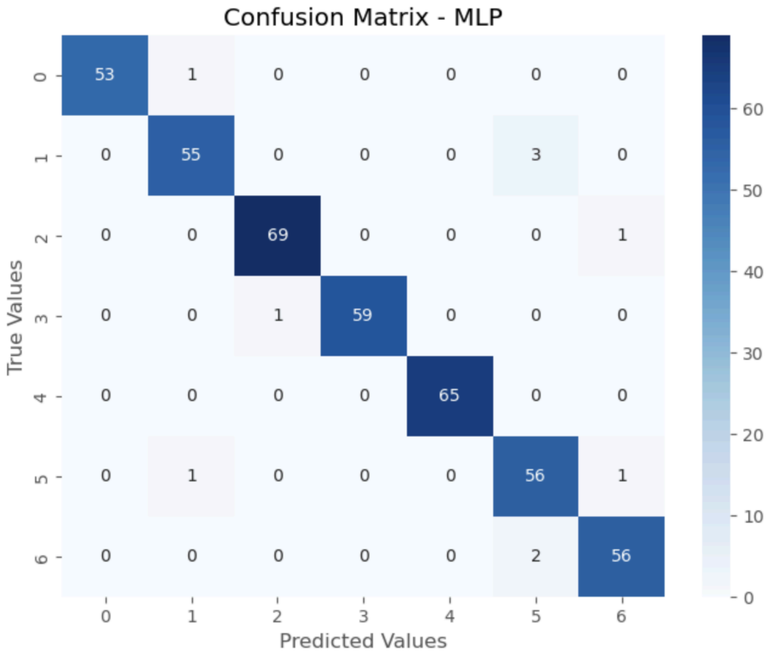

Figure 19: Confusion Matrix of Model 4

The confusion matrix further supports these findings, showing minimal misclassifications. Most predictions aligned accurately with true labels, though a few instances of confusion appeared between closely related classes, such as between classes 1 and 5. This suggests the model effectively distinguishes among categories, with only minor overlaps that could be explored for further refinement.

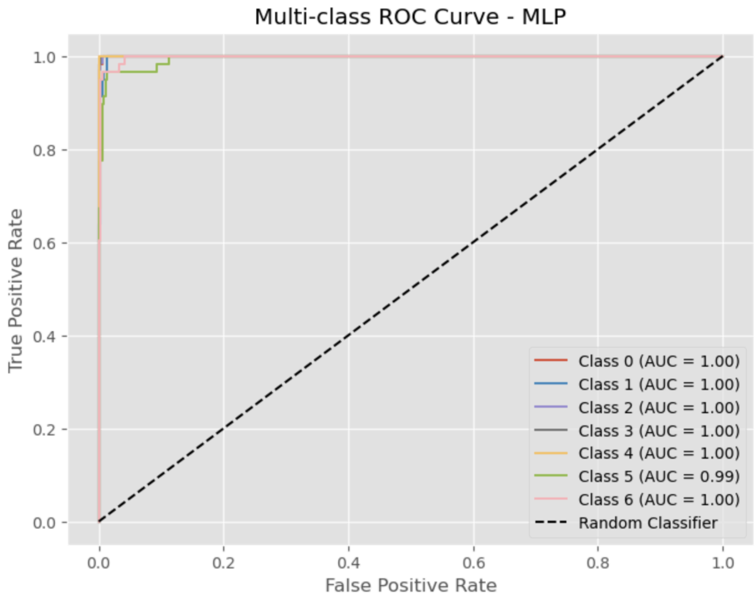

Figure 20: ROC Curve of Model 4

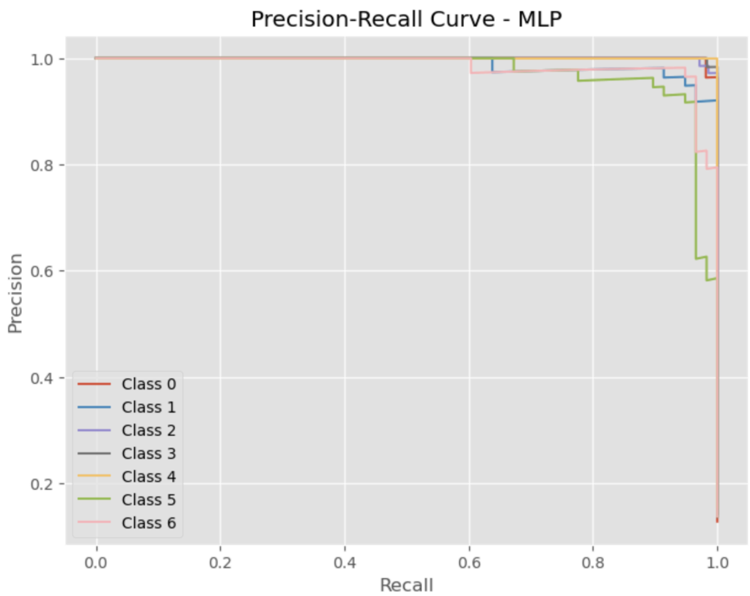

Figure 21: Precision-Recall Curve of Model 4

Additionally, the multi-class ROC curves demonstrated excellent discriminative capability, with AUC values of 1.00 for nearly all classes and 0.99 for class 5, signifying high sensitivity and specificity. The precision-recall curves complemented these insights, showing that precision remained high even as recall increased, particularly for the more challenging classes.

The evaluation demonstrates that the neural network model was both robust and accurate, with performance metrics and visualizations confirming its suitability for multi-class classification tasks.

7.5 Model 5: Deep Neural Network with Multi-Layer Perceptron

Model 5 was built using a deep neural network architecture implemented through Keras, following a Multi-Layer Perceptron (MLP) structure. This model was composed of multiple densely connected layers, allowing it to learn complex, non-linear relationships among the input features. The choice to use a deep learning model was driven by its ability to model intricate feature interactions that simpler models might miss. In the context of this multi-class classification task, a neural network was expected to extract meaningful patterns and improve prediction accuracy through its depth and flexibility.

To refine the model's structure and performance, hyperparameter tuning was conducted through a grid search strategy combined with 5-fold stratified cross-validation. This tuning effort explored different configurations of hidden layer sizes, activation functions, dropout rates, and learning rates. After evaluating 24 combinations, the optimal model structure was found to consist of three hidden layers with sizes of 256, 128, and 64 neurons, respectively. The ReLU activation function was used, with a dropout rate of 0.3 to mitigate overfitting, and the learning rate was set at 0.001.

Following this optimization, the model was retrained and tested, yielding an updated accuracy of 93.38%.The tuned model's accuracy was lower than that of Model 4.

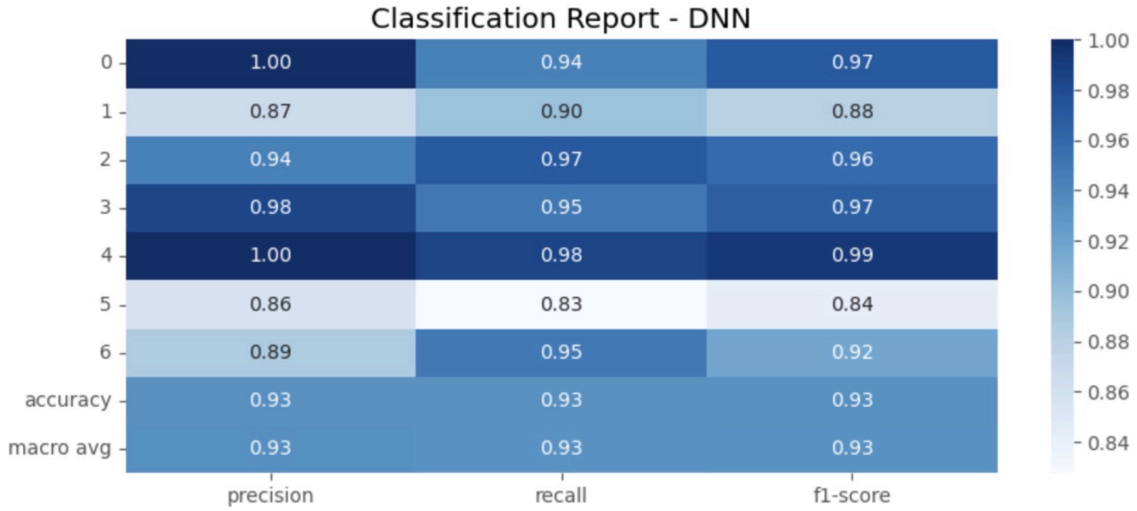

Figure 22: Classification Report of Model 5

The classification report reveals an overall accuracy of 93%, with a macro-averaged F1-score also at 0.93, indicating consistent predictive strength across all classes. Class-specific performance varies, with precision reaching 100% for class 0 and class 4, while class 5 shows comparatively lower precision and recall. The model's recall is strongest for class 4 and class 2, suggesting robust detection in those categories, although class 5 again stands out with a slightly reduced recall of 0.83, indicating some difficulty in capturing all relevant instances for that group.

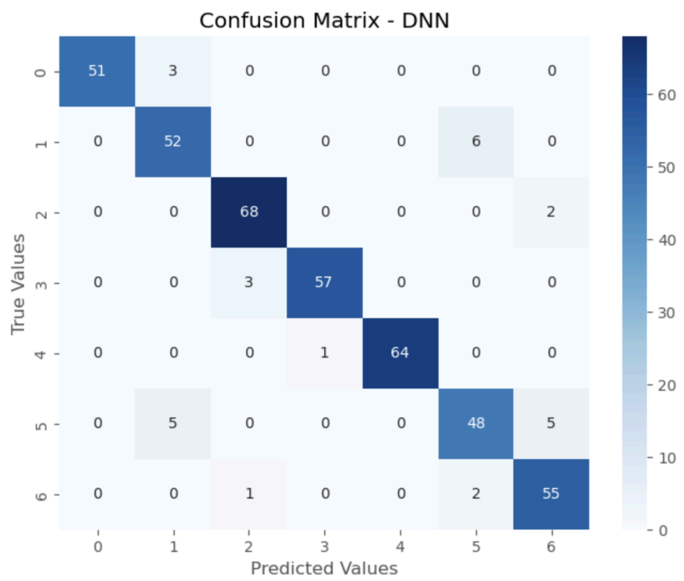

Figure 23: Confusion Matrix of Model 5

The confusion matrix highlights this variability, with most classes displaying minimal misclassification. However, there is notable confusion between classes 1 and 5, and classes 5 and 6, as evidenced by misassigned predictions. These misclassifications help identify potential overlaps in feature representation or areas where additional data preprocessing or rebalancing might improve clarity between labels.

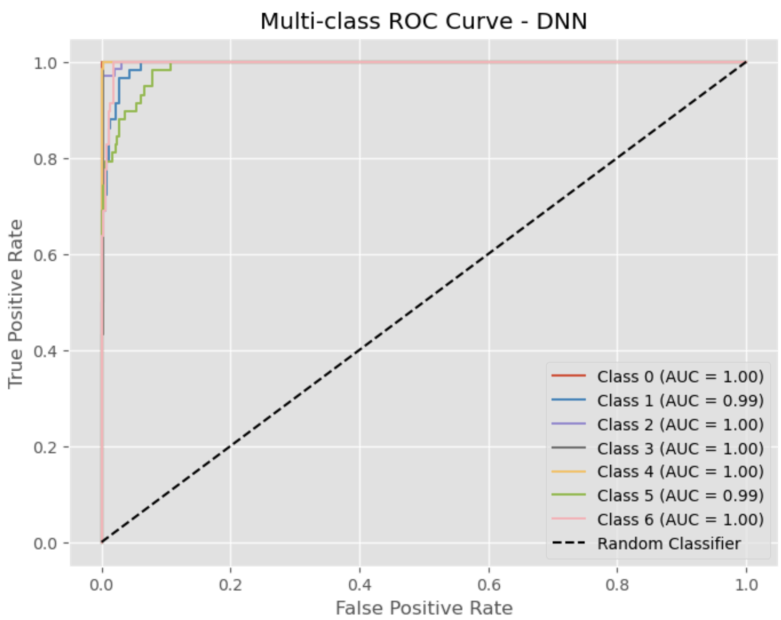

Figure 24: ROC Curve of Model 5

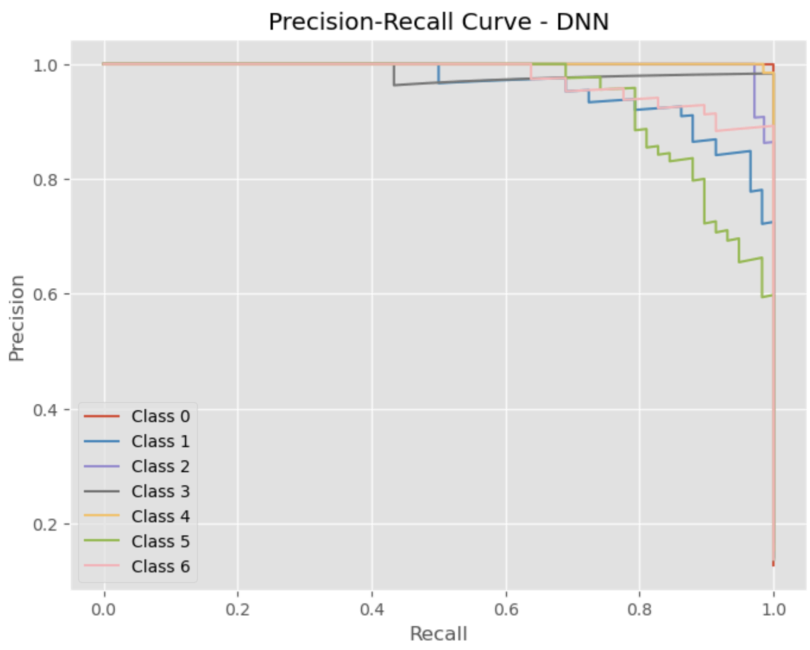

Figure 25: Precision-Recall Curve of Model 5

The multi-class ROC curve further supports the model's effectiveness, showing AUC scores of 1.00 for most classes and 0.99 for classes 1 and 5, reflecting high discriminatory power. Similarly, the precision-recall curve illustrates stable precision across increasing recall values for most classes, though precision declines slightly at higher recall levels for class 5 and class 6. These curves confirm that the model maintains a balance between sensitivity and specificity across diverse class distributions.

Although the model demonstrated solid accuracy, its overall results were somewhat lower than those of Model 4, suggesting that while deep learning provided valuable insights, its complexity did not translate into superior performance in this case. These findings underscore the importance of aligning model complexity with dataset characteristics for optimal outcomes.

7.6 Comparison & Selection

If model transparency and interpretability are primary concerns, Random Forest and Gradient Boosting stand out as suitable options. Both models not only deliver strong predictive accuracy but also provide insight into feature importance, enabling a clearer understanding of which variables contribute most to classification outcomes. This level of interpretability can be especially valuable when model decisions must be explained to stakeholders or used in settings where transparency is critical.

In contrast, when the goal is to achieve the highest possible classification accuracy, Support Vector Machine (SVM) and scikit-learn's Multi-Layer Perceptron (MLP) offer superior performance. These models produced the most precise classifications in the current dataset, with MLP achieving the highest test accuracy among all models. Both handled the multi-class classification task effectively and demonstrated robust performance across challenging classes. However, they do so at the expense of interpretability, particularly in the case of MLP, where understanding internal decision mechanisms is more complex.

Deep Neural Network (DNN), although designed to handle complex non-linear patterns, showed a slight decline in performance relative to SVM and MLP. Given the relatively small size of the dataset, the DNN appeared to overfit, leading to a reduction in generalization. This model type typically excels when applied to larger datasets, where its capacity to learn intricate patterns can be fully leveraged. In this context, the DNN's complexity did not translate into a significant performance advantage and was outperformed by more streamlined models.

Overall, all five models performed well, but their suitability depends on specific priorities. Random Forest and Gradient Boosting offer a balance between accuracy and interpretability, SVM and MLP prioritize precision, while DNN holds potential for scalability in larger datasets but may not be optimal for smaller ones. Selecting the most appropriate model requires weighing the trade-offs between performance, interpretability, and dataset size.

8.0 Model Deployment Strategy

The model deployment strategy focuses on ensuring data security and privacy while maintaining a streamlined and automated machine learning pipeline. Given that the dataset includes sensitive medical and personal information, deploying the model in a cloud environment or using API-based solutions was deemed unsuitable due to potential security risks. As a result, an on-premises deployment approach was selected to keep all data storage, processing, and model execution within a secure local infrastructure.

Automation plays a significant role throughout the ML pipeline, starting from data ingestion and preprocessing to model training and evaluation. Data is collected and updated through automated scripts, and preprocessing steps such as cleaning, feature engineering, and normalization are integrated into a reproducible pipeline using tools like scikit-learn and Python-based workflow managers. Once new data becomes available, the pipeline can be re-triggered to perform incremental training, allowing the model to adapt to updated information without requiring full retraining from scratch.

Model optimization, including hyperparameter tuning and cross-validation, is incorporated into the automated pipeline, ensuring that every version of the model is rigorously tested before being moved into production. Evaluation metrics such as accuracy, F1 score, and ROC-AUC are logged and monitored to assess ongoing performance.

Once deployed locally, the model is integrated into the existing data systems for real-time or batch predictions, depending on operational needs. Maintenance in production includes regular performance monitoring to detect drift or degradation, with automated alerts prompting retraining or manual review if predefined thresholds are exceeded. This strategy ensures that the model remains accurate and secure over time while complying with data privacy requirements.

9.0 Findings

Obesity is one of the most pressing public health challenges worldwide, with its prevalence rising rapidly across both developed and developing countries. Beyond its impact on body weight, obesity is a major risk factor for type 2 diabetes, hypertension, cardiovascular diseases, liver disorders, and even mental health conditions like depression. The economic burden is equally severe, particularly in low-income regions where access to healthcare is limited. While developed nations have historically reported higher obesity rates, recent trends show a steeper rise in developing countries such as Mexico, Brazil, and Colombia, where urbanization, processed food consumption, and sedentary lifestyles are fueling the crisis. Given these alarming trends, early identification of obesity risk factors and targeted intervention strategies are essential.

Our study aimed to explore the relationships between demographic factors, dietary habits, physical activity levels, and obesity, using machine learning models to identify key predictors and generate actionable insights. Through data preprocessing and feature selection, we found that high-calorie food consumption, family history of obesity, weight, height, and physical activity frequency were among the strongest predictors. Interestingly, screen time and hydration levels had minimal impact, suggesting that some lifestyle choices may play a smaller role in obesity development than commonly assumed.

When evaluating model performance, MLP (Multi-Layer Perceptron) consistently outperformed all other models, achieving the highest accuracy and maintaining a strong balance between precision, recall, and F1-score. SVM and Gradient Boosting also performed well, making them strong alternatives. However, Random Forest struggled slightly in classifying minority obesity categories, and DNN (Deep Neural Network) surprisingly underperformed. This likely reflects the dataset's limited size, which may have prevented deep learning models from fully leveraging their predictive capabilities. This finding aligns with broader research suggesting that more complex models aren't always better, and well-tuned traditional models like MLP and SVM can sometimes outperform deep learning in structured health-related data.

Additionally, our models achieved high accuracy across all evaluation metrics, which can be attributed to a well-structured dataset with clearly defined relationships between features and the target variable. The balanced class distribution also helped prevent bias, ensuring that no single category was overrepresented. However, high accuracy does not always guarantee real-world reliability. Some models, particularly deep learning-based ones, may have overfit the dataset, capturing patterns that work well in this specific data but may not generalize to new populations. Since obesity trends vary significantly across regions, income levels, and healthcare access, models trained on this dataset may not perform as well in different settings.

10.0 Conclusions

Obesity is not just a medical condition—it is a complex, multifaceted issue influenced by genetics, lifestyle, environment, and access to healthcare. This study demonstrates that machine learning can be a powerful asset in predicting obesity risk, allowing for early detection and data-driven interventions. Based on all evaluation metrics, MLP emerged as the best-performing model, offering the highest accuracy and generalizability. SVM and Gradient Boosting also performed well, making them strong secondary options for predictive modeling.

One of the key takeaways from this study is that deep learning models like DNN did not outperform simpler methods. While deep learning has revolutionized many fields, this finding reinforces that traditional machine learning models, when properly tuned, can be just as effective for structured, tabular data like health records. The importance of quality data, feature selection, and interpretability cannot be overstated—without the right predictors, even the most sophisticated models will struggle to generate meaningful insights.

However, predictive models alone will not solve the obesity epidemic. The real challenge lies in translating data-driven insights into actionable public health strategies. Governments must focus on preventative healthcare, improving food accessibility, and encouraging physical activity, particularly in low-income and urban communities where obesity rates are rising the fastest. By combining machine learning, policy interventions, and behavioral change initiatives, we can work toward a future where obesity rates decline, healthcare costs are reduced, and people live healthier, longer lives.

11.0 Limitations

Even though our models performed well overall, there are a few important limitations that should be acknowledged.

First, the dataset size and diversity could be a concern. While our models achieved high accuracy on the test set, the dataset primarily focused on dietary habits and physical activity, which, although crucial, do not fully capture the complexity of obesity. Other influential factors, such as socioeconomic status, healthcare access, and genetic predisposition, were not included, which may limit the model's generalizability. Additionally, since obesity trends vary significantly across different regions, age groups, and income levels, a more diverse dataset would help ensure that the model is applicable to a broader population.

Second, our feature selection process had limitations. We selected features based on initial data exploration, but we did not fully explore advanced feature engineering techniques such as interaction terms, polynomial transformations, or automated feature selection. Given that obesity is influenced by a combination of behavioral, genetic, and environmental factors, more sophisticated feature extraction methods could have helped identify hidden patterns and improve model performance.

Another limitation is related to model selection and hyperparameter tuning. We used grid search to optimize hyperparameters, but this method can be computationally expensive and may not always find the absolute best configuration. More advanced techniques like Bayesian Optimization or Random Search could have provided better tuning results in a shorter time. Additionally, while we tested five machine learning models, there may be other approaches—such as AutoML frameworks or transformer-based models—that could further improve prediction accuracy.

When it comes to model evaluation, we relied on metrics like accuracy, precision, recall, F1-score, ROC-AUC, and PR curves, which are standard for classification tasks. However, real-world applications often require additional considerations, such as computational efficiency, inference speed, and interpretability. For instance, deep learning models like DNN and MLP, while highly accurate, are more computationally expensive than traditional models like SVM and Gradient Boosting. This could pose a challenge if the model were to be deployed in real-time healthcare applications or resource-limited environments.

Lastly, potential data biases and noise were not deeply examined. If the dataset contains measurement errors, survey biases, or underrepresented subpopulations, the model's predictions could be skewed. In real-world applications, improving data quality through outlier detection, data augmentation, and integrating additional health data sources (such as medical history or wearable device data) would likely make the model more reliable.

Overall, while our models provide valuable insights into obesity risk prediction, addressing these limitations—expanding the dataset, refining feature selection, exploring alternative models, and improving evaluation strategies—could further enhance their accuracy, interpretability, and real-world applicability.

12.0 Recommendations

Obesity is not just a health condition—it's a growing public health, economic, and social crisis that requires urgent intervention. While our study used machine learning to predict obesity risk, the real challenge lies in turning these predictions into meaningful actions. Effective solutions must go beyond individual lifestyle changes and include policy reforms, better healthcare access, and advancements in technology to create long-term impact.

One of the biggest takeaways from this study is that obesity prevention efforts need to be more comprehensive. Public health policies should prioritize early detection and targeted interventions, using data-driven approaches to identify high-risk groups. For example, governments could improve school nutrition programs, impose stricter regulations on processed food marketing, and provide incentives for healthier eating habits—especially in low-income communities where obesity rates are rising the fastest. Countries like Mexico and Brazil have experimented with sugar taxes and front-of-package warning labels, and expanding these initiatives globally could help slow down obesity trends. However, policy alone isn't enough—access to affordable, nutritious food and safe spaces for physical activity must also be improved, particularly in developing countries where urbanization has made active lifestyles harder to maintain.

Technology can also play a major role in personalized obesity prevention and management. AI-powered health apps could provide real-time, tailored guidance on diet and exercise, helping individuals make more informed lifestyle choices. Wearable devices and mobile tracking tools could be integrated with obesity risk prediction models to offer continuous monitoring and instant feedback, empowering users to stay on track. Telehealth and virtual nutrition counseling could help bridge the gap for individuals who lack access to healthcare professionals, ensuring that more people receive preventative care before obesity-related complications develop.

From a broader perspective, economic factors must also be addressed. Treating obesity-related diseases is financially unsustainable, particularly in developing countries with limited healthcare infrastructure. Governments and international health organizations must invest in long-term prevention strategies, rather than simply managing the consequences. More funding for public health research, stronger regulations on ultra-processed food industries, and better healthcare infrastructure will be critical in the years ahead.

Ultimately, no single approach will solve the obesity epidemic. It will take a combination of smarter policies, better healthcare access, and technology-driven solutions to make real progress. But one thing is clear: waiting until obesity rates surge even higher is not an option. Now is the time to act—through data, policy, and innovation—to create a healthier future for the next generations.

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